

PROJECT-2

INPUT

```
GNU nano 8.7.1 sample.txt *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int source_fd, destination_fd;
    char buffer[1024];
    ssize_t bytes_read, bytes_written;

    // Open source file for reading
    source_fd = open("source.txt", O_RDONLY);

    if (source_fd < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    // Open destination file for writing
    // Create it if it does not exist
    // Truncate it if it already exists
    destination_fd = open("destination.txt",
                        O_WRONLY | O_CREAT | O_TRUNC,
                        0644);

    if (destination_fd < 0)
    {
        perror("Error opening destination file");
        close(source_fd);
        return 1;
    }

    // Read from source and write to destination
    while ((bytes_read = read(source_fd, buffer, sizeof(buffer))) > 0)
    {

```

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                                0644);

        if (destination_fd < 0)
        {
            perror("Error opening destination file");
            close(source_fd);
            return 1;
        }

        // Read from source and write to destination
        while ((bytes_read = read(source_fd, buffer, sizeof(buffer))) > 0)
        {
            bytes_written = write(destination_fd, buffer, bytes_read);

            if (bytes_written != bytes_read)
            {
                perror("Error writing to destination file");
                close(source_fd);
                close(destination_fd);
                return 1;
            }
        }

        if (bytes_read < 0)
        {
            perror("Error reading source file");
        }

        // Close both files
        close(source_fd);
        close(destination_fd);

        printf("File copied successfully.\n");

        return 0;
    }
}
```

OUTPUT

```
administrator@VBCO-LAP-29: ~$ mkdir OSLab
mkdir: OSLab: File exists
administrator@VBCO-LAP-29: ~$ cd fork
-bash: cd: fork: No such file or directory
administrator@VBCO-LAP-29: ~$ cd OSLab
administrator@VBCO-LAP-29: ~/OSLab$ cd fork
administrator@VBCO-LAP-29: ~/OSLab/fork$ nano practical02.c
administrator@VBCO-LAP-29: ~/OSLab/fork$ nano sample.txt
administrator@VBCO-LAP-29: ~/OSLab/fork$ gcc practical02.c -o practical02
administrator@VBCO-LAP-29: ~/OSLab/fork$ ./practical02
Error opening source file: No such file or directory
administrator@VBCO-LAP-29: ~/OSLab/fork$ ../practical02
Error opening source file: No such file or directory
administrator@VBCO-LAP-29: ~/OSLab/fork$ ls
OSLab  Ques.1  practical02  practical1  pratical01.c  pro1.c  pro2.c  pro3.c  sample.txt
Pro1.c  Ques.1.c  practical02.c  practical01  pro1          pro2      pro3      pro4.c
administrator@VBCO-LAP-29: ~/OSLab/fork$ mv sample.txt source.txt
administrator@VBCO-LAP-29: ~/OSLab/fork$ ./practical02
```

```
Operating Systems Lab
Using open(), read(), write(), close().
This file is copied using system calls.
```